MILITARY SPECIFICATION

DELAY ELEMENT, T6E4 AND T5E3 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for two types of delay elements designated as Delay Element, T6E4 and T5E3 (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR–S–366 —Sieves, Standard for Testing Purposes.

MILITARY

MIL–P–116 —Preservation, Methods of.

MIL–A–2550 —Ammunition and Special Weapons; General Specification for.

MIL–P–20444 —Primer, Percussion, M42, Loading, Assembling and Packing.

MIL–I–45208 —Inspection Requirements; General Specification for.

MIL–I–45607 —Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 —Quality Assurance Terms and Definitions.

MIL–STD–1234 —Pyrotechnic Sampling, Inspection and Testing.

MIL–STD–1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8822519 —Delay Element, T6E4 and T5E3, Assembly.

8846861 —Box, Packing, Ammunition, for Delay Elements, T5E3 or T6E4.

8846862 —Carton, Packing, Ammunition, for Delay Elements, T5E3 or T6E4.

PUBLICATIONS

ORDNANCE CORPS

IEL—8822519 —Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Elements. The loaded delay element assemblies shall comply with all requirements specified on Drawing (dwg.) 8822519, and with all requirements specified in applicable specifications.

3.3 Moisture content.

3.3.1 *Igniter and flash composition.* The moisture content of igniter and flash composition at the time of loading at the loading station shall be not greater than 0.05 percent when determined as specified in 4.3.1.

3.3.2 *Delay composition.* The moisture content of delay composition at the time of loading shall be not greater than 0.04 percent when determined as specified in 4.3.1.

3.4 Water soluble chloride content. The water soluble chloride content of the igniter and flash or delay composition at the time of loading shall be not greater than 0.05 percent when determined as specified in 4.3.2.

3.5 Static functioning. Flame is permitted at the vent hole, only. Testing shall be as specified in 4.3.5.

3.5.1 *Delay element, T5E3.* The T5E3 element shall be considered to have functioned properly, if the elapsed time between the initiation of the primer and completion of functioning is between four seconds and eight seconds.

3.5.2 *Delay element, T6E4.* The T6E4 element shall be considered to have functioned properly, if the elapsed time between the initiation of the primer and completion of functioning is between eleven seconds and seventeen seconds.

3.6 Impact functioning. Delay element assemblies, following temperature conditioning, shall not fail to function and shall comply with the applicable delay requirement in table I when subjected to the impact functioning test as specified in 4.3.6. All temperatures shall be in degrees Fahrenheit (F.).

3.7 Workmanship. All manufacturing, process and assembly shall be performed in a thorough, workmanlike manner. Parts shall be dry, free or burrs, chips, dirt, grease, rust, or other foreign material. The cleaning method shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Protective coatings shall be continuous except for a few slight scratches not exposing base material. All required markings shall be neat and sharply defined.

TABLE I.

Element	Temperature	Delay time seconds (sec)	
		Maximum	Minimum
T6E4.....	160, plus or minus 3.....	17	11
	Minus 65, plus or minus 3....	17	11
T5E3.....	160, plus or minus 3.....	8	4
	Minus 65, plus or minus 3....	8	4

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contrac-

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tor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analyses on all material purchased directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of delay element assemblies of one type produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Parts bearing the same lot interfix number from one manufacturer.

(b) Delay composition from one batch or lot from one manufacturer.

(c) Igniter and flash composition from one batch or lot from one manufacturer.

(d) Primers bearing the same lot interfix number from one manufacturer.

(e) Relays bearing the same lot interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Body assembly, prior to crimping disc.* (See dwg. 8822520 covering a detail of dwg. 8822519.)

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Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Vent hole obstructed	Visual	01001
202.	Sealing material missing on disc	Visual	01002
203.	Evidence of poor workmanship (see 3.7)	Visual	01003

4.2.2.2 *Loading assembly, body.* (See dwg. 8822521 covering a detail of dwg. 8822519.)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge in primer cavity	Visual	02001
102.	Blow through hole obstructed	Visual	02002
103.	Charge missing	Visual	02003
Minor:	None defined.		

4.2.2.3 *Assembly.* (See dwg. 8822519.)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Primer missing	Visual	03001
102.	Primer deformed or ruptured due to crimping	Visual	03002
103.	Vent hole disc ruptured	Visual	03003
104.	Primer not crimped full 360 degrees	Visual	03004
105.	Plug stake missing	Visual	03005
106.	Excessive sealing compound on primer	Visual	03006
Minor:	AQL 2.5 percent		
201.	Length, max.	Gage	03007
202.	Width, max.	Gage	03008
203.	Thickness, max.	Gage	03009
204.	Crimp of vent hole disc incomplete	Visual	03010
205.	Excessive sealing compound on plug or on vent hole disc	Visual	03011
206.	Marking incorrect, misleading or unidentifiable	Visual	03012
207.	Evidence of poor workmanship (see 3.7)	Visual	03013

4.2.2.4 *Carton, prior to sealing.* (See dwg. 8846862.)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Packing component missing	Visual	04001
Minor:	AQL 0.40 percent		
201.	Separator damaged to extent that items touch each other	Visual	04002

4.2.2.5 *Carton, sealed.* (See dwg. 8846862.)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Sealing strip torn, badly wrinkled or incomplete	Visual	05001
202.	Carton torn, cut, or punctured	Visual	05002
203.	Contents loose	Manual	05003
204.	Marking incorrect, misleading, or unidentifiable	Visual	05004

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4.2.2.6 *Sealed barrier bag.* (See dwg. 8846862.)

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Seal improper or incomplete	Visual	06001
102.	Bag punctured, torn, or cut	Visual	06002
103.	Seal not permitting minimum subsequent seals	Visual	06003
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading, or unidentifiable	Visual	06004

4.2.2.7 *Sealed wood packing box.* (See dwg. 8846861.)

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Hardware missing, broken or loose	Visual-Manual	07001
102.	Strapping missing, broken, or loose	Visual-Manual	07002
103.	Box damaged (contents exposed or liable to be exposed)	Visual	07003
104.	ICC nomenclature incorrect misleading, or unidentifiable	Visual	07004
Minor:	AQL 2.5 percent		
201.	Hardware improperly engaged	Visual	07005
202.	Strapping improperly assembled	Visual	07006
203.	Car seal unsealed or improperly positioned	Visual	07007
204.	Marking incorrect, misleading or unidentifiable	Visual	07008
205.	Contents loose when tilted	Visual-Manual	07009

4.2.3 *Testing.*

4.2.3.1 *Igniter and flash composition; or delay composition.* A sample of fifty (50) grams shall be selected from each lot or batch of each composition to perform the following analyses:

4.2.3.1.1 *Determination of igniter and flash composition (see dwg. 8822521)—Major defect (Code No. 02004); or of delay composition—Major defect (Code No. 02005).* A portion of four (4) grams of the sample of 4.2.3.1 shall be selected for these analyses which shall be performed as specified in 4.3.3 or 4.3.4, as applicable. Failure of a portion in meeting one or more of the requirements shall reject the product represented by the sample.

4.2.3.1.2 *Determination of moisture content, igniter and flash composition (see 3.3.1) Code No. 02006 or delay composition (see 3.3.2) Code No. 02007—Major defect.* A portion of twenty (20) grams shall be transferred from the applicable sample of 4.2.3.1 for each determination. The determination shall be performed as specified in 4.3.1. Failure of a portion in meeting the requirements shall reject the product represented by the sample.

4.2.3.1.3 *Determination of water soluble chloride content (see 3.4) igniter and flash composition—Code No. 02008 or delay composition—Code No. 02009—Major defects.* A portion of twenty (20) grams shall be transferred from the applicable sample of 4.2.3.1 for each determination. The determination shall be performed as specified in 4.3.2. Failure of a portion in meeting the requirement shall reject the product represented by the sample.

4.2.3.2 *Static functioning (see 3.5.1) Code No. 03014 or (see 3.5.2) Code No. 03015—Major defects.* A sample of 50 units shall be selected from each lot for this test which shall be performed as specified in 4.3.5 using the applicable equipment of 4.2.4. Failure of one or more units in meeting the applicable requirement shall reject the lot.

4.2.3.2.1 *Static functioning.* After three

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consecutive lots have complied with the criteria of 4.2.3.2, no further static functioning tests shall be performed.

4.2.3.3 *Impact functioning test (see 3.6).*

4.2.3.3.1 *At 160 degrees F. T5E3—Code No. 03016; T6E4—Code No. 03017—Major defects.* A sample of 50 units shall be selected from each lot for high temperature testing which shall be performed as specified in 4.3.6 using the applicable equipment of 4.2.4. Failure of 2 or more units in meeting the applicable requirement shall reject the lot.

4.2.3.3.2 *At minus 65 degrees F. T5E3—Code No. 03018; T6E4—Code No. 03019—Major defects.* The sample size, test procedure and rejection criteria for low temperature testing shall be the same as those for 4.2.3.3.1.

4.2.3.4 *Heat seal or quick leak test of barrier bag, as applicable (see dwg. 8846862)—Major defect—Code No. 06005.* The test procedures for the test shall be as prescribed in Specification MIL–P–116. Sampling and acceptance criteria shall be in accordance with inspection level S–4 of Standard MIL–STD–105.

Note. The test detailed in paragraph 4.4.3.2.1 of Specification MIL–P–116 shall not be used.

4.2.3.5 *Check test for primer deterioration.* If the total time between original acceptance of the primer lot and the assembly of the lot into the delay element exceeds two years or if the primers have been subjected to adverse conditions, however brief, at any time since the last previous test, the primers shall be subjected to, and must satisfactorily pass, the check test specified in the applicable primer loading specification MIL–P–20444 before the primers are assembled into the Delay Element (see 6.5).

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL–8822519 identifies the inspection equipment required to perform the examination and tests prescribed in this section. The contractor shall design inspection equipment in accordance with 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of moisture content of igniter and flash composition (see 3.3.1), and of delay composition (see 3.3.2).* An accurately weighed portion of 10 grams shall be taken from the sample in 4.2.3.1.2 for this test. Test shall be conducted in accordance with Method 102.1 of MIL–STD–1234. Determination shall be made for conformance with 3.3 (as applicable.)

4.3.2 *Determination of water soluble chloride content of delay charge composition, or of igniter composition (see 3.4).* Dissolve 1 gram from 4.2.3.1.3 in about 60 ml. of hot distilled water. Cool the solution to room temperature, transfer to 100 ml. volumetric flask and dilute to volume. Pipette 10 ml. of this solution into a Nessler tube, and dilute to 30 ml. with distilled water. Add 30 ml. of distilled water to another Nessler tube. Add 1 ml. concentrated HNO3 and 1 ml. of 0.1N AgNO3 solution to the sample and blank. To the blank, add a sufficient amount of sodium chloride

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solution (0.825g/1) that will match the turbidity of the sample. Allow the solutions to stand 15 minutes in the dark, then, visually compare the sample with the standard. After the turbidities are matched the percent chlorides shall be calculated and compared with the requirements.

Calculation:

$$\text{Percent of chlorides} = .50 \frac{V}{Va}$$

where:

V = volume of standard sodium chloride solution required to match or exceed the turbidity of the sample.

Va = volume of solution used.

4.3.3 *Determination of igniter and flash composition (see dwg. 8822521).*

4.3.3.1 *Diatomaceous earth (silicon dioxide).* Transfer a one-gram sample from 4.2.3.1.1 to a 250 ml. Erlenmeyer flask, and add 20 ml. of 85 percent sulfuric acid (specific gravity 1.77). Heat gently at first to dissolve the ferric oxide and then more vigorously to dissolve the zirconium. Continue the heating until fumes of sulfuric acid are evolved. Ferric sulfate will separate from the solution. Cool, and cautiously dilute with 50 ml. of water to dissove the salts. If any red or gray particles remain undissolved, again heat to fumes of sulfuric acid. Cool and dilute to 150 ml. with water. Filter through a No. 42 Whatman filter paper (or its equivalent), and wash with about 40 ml. of warm sulfuric acid (1:100). Collect the filtrate in a 250 ml. volumetric flask. Place the filter paper in a platinum crucible, and dry in an oven at 100 degrees to 110 degrees Centigrade (C.) for 2 hours. Then place the crucible and its contents in a cold muffle furnace, and raise the temperature to read 300 to 400 degrees C. until the paper has been completely charred. Raise the temperature to 1000 degrees plus or minus 50 degrees C. and ignite for ½ hour. Cool in a desiccator, and weigh. To the ignited residue, add 5 ml. of concentrated hydrofluoric acid and one drop of concentrated sulfuric acid. Evaporate the acid, ignite and reweigh. The difference in weight before and after adding the hydrofluoric acid is due to the loss of silica and this difference should be inserted in the numerator of the following equation:

$$\text{Percentage of silicon dioxide} = \frac{(\text{loss in weight}) \ 100}{\text{Weight of sample.}}$$

Fuze the volatile residue with one grain of potassium bisulfate. Dissolve the cooled melt in hot water, and add the solution to the 250 ml. flask that contains the silica filtrate. Dilute to the mark, mix, and reserve for determination of iron and zirconium.

4.3.3.2 *Iron (III) oxide.* Take a 50 ml. aliquot from the 250 ml. flask containing the silica filtrate in 4.3.3.1 and add 50 ml. of concentrated hydrochloric acid. Heat the solution nearly to boiling and add stannous chloride solution (3 grams SnC12 dissolved in 10 ml. hydrochloric acid diluted to 20 ml. with water) dropwise until the color of the of the solution changes from yellow to light green. If this color change is not sharp, discard the solution, take a new aliquot, evaporate to 10–15 ml., add an equal volume of concentrated hydrochloric acid and proceed with the stannous chloride reduction as described above. Add one to two drops of stannous chloride in excess, cool to room temperature and rapidly add 10 ml. of mercuric chloride solution (saturated aqueous solution of HgC12). If the precipitate formed is very heavy or is other than white, too much stannous chloride has been used and the reduction process must be repeated. Allow the solution to stand for about five minutes and then transfer to a 600 ml. beaker with the aid of about 250 ml. of water. Cautiously add 15 ml. of sulfuric acid, five ml. of phosphoric acid, and 6 to 8 drops of sodium diphenylamine sulfonate indicator solution (0.2 gm/100 ml of water). Titrate slowly with 0.05N standard potassium dichromate solution until the pure green color changes to a gray-green. Then add the dichromate one drop at a time until the first tinge of purple or violet-blue appears.

Percentage of iron (III) oxide = 39.92VN divided by W

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where:

V = ml. of standard potassium dichromate solution used.
N = normality of standard potassium dichromate solution.
W = weight of sample in grams.

4.3.3.3 *Zirconium (see 6.9).* Transfer a 50 ml. aliquot of the silica filtrate from 4.3.3.1 to a 250 ml. beaker. Add 50 ml. of mandelic acid solution (16 grams in 100 ml. of water), and place on a steam bath for one hour. Remove from the steam bath and let stand for at least 3 hours. Filter the solution through a 11 cm. No. 40 Whatman filter paper (or its equivalent) and wash the precipitate with a solution containing 5 grams of mandelic acid and 2 ml. of hydrochloric acid per 100 ml. of water. Use a policeman to remove the precipitate adhering to the stirring rod and the beaker. Place the filter paper in a weighed platinum crucible, and dry in an oven at 100 to 110 degrees C. for 2 hours. Then place the crucible and its contents in a cold muffle furnace, and raise the temperature to 300 to 400 degrees C. until the paper has been completely charred. Raise the temperature to 1000 plus or minus 50 degrees C. and ignite for ½ hour. Cool in a desiccator and weigh as zirconium oxide (ZrO_2). However, if the residue is brown, add 1 gram of potassium bisulfate, mix well, and fuze. Cool, dissolve in hot water, add 5 ml. of concentrated hydrochloric acid, dilute to 50 ml., and proceed as before by precipitating with mandelic acid.

Percentage of zirconium = (Weight of ZrO_2) 370.2 divided by Weight of sample.

4.3.4 *Determination of delay composition.*

4.3.4.1 *Lead chromate or barium chromate.* Transfer a 0.5 gram sample from the material in 4.2.3.1.1 and dissolve it in 200 ml. of 10 percent hydrochloric acid in a 500 ml. covered Erlenmeyer flask. Heat to boiling and filter through Whatman No. 40 filter paper. Wash the paper four times with one percent hydrochloric acid and combine washings with filtrate. Heat the solution to boiling and add 20 ml. of 50 percent sulfuric acid slowly. Continue boiling for 5 minutes and allow to stand at 60 degrees C. or higher until the liquid is clear. Decant through a tared, fine porosity, fritted porcelain crucible and transfer the precipitate to the filter. Wash three times with 0.5 percent sulfuric acid and then 50 percent alcohol until free of acid. Dry the precipitate for 30 minutes at 110 degrees C. and then ignite for 60 minutes at 750 degrees C. Weigh and record as barium sulfate and lead sulfate. Dissolve the lead sulfate from the crucible by extraction with hot ammonium acetate solution (20 grams ammonium acetate and 3 ml. 80 percent acetic acid per 100 ml. of water). Wash thoroughly with hot water.

Wash the residue of the sulfate with hot 10 percent sulfuric acid to remove the acetate, then with 50 percent alcohol until free of acid. Dry the precipitate as before. Weigh and record as barium sulfate. Determine percentage of initial sample and multiply the percentage by 1.085 to convert to percent barium chromate. The weight of the lead sulfate and barium sulfate less the weight of the barium sulfate is equal to the weight of the lead sulfate. Determine percentage of initial sample and multiply this percentage by 1.06 to convert to percent lead chromate.

4.3.4.2 *Manganese, powdered.* Transfer a 0.500 gram sample of material from the sample of material in 4.2.3.1.1 to a 500 ml. Erlenmeyer flask and cover with 25 ml. of water. Dissolve in 30 ml. of 50 percent nitric acid and 2 ml. of 600 percent perchloric acid. Evaporate the solution until almost dry, and then cool. Add 50 ml. of water, 1 ml. of 10 percent ferric nitrate and 2.0 percent of potassium chromate. Cover the flask with a watch glass and boil the solution for 15 minutes. Cool and adjust the volume to 50 ml. Filter through a Whatman No. 42 filter paper and ignite to manganese dioxide (MnO_2) in a tared crucible. Dissolve manganese dioxide in sulfuric acid, and expel the excess acid by heating. Filter, wash with

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water, dry the residue and weigh as manganese sulfate. Determine the percentage of initial sample and multiply this percentage by 0.3638 to convert to percent manganese in required mixture.

4.3.5 *Static functioning.* The element shall be assembled into the applicable test equipment. (The equipment and lighting condition shall be such that flame emitting from the element during test can be observed.) The element shall be functioned. Any unit which exhibits flame at any point other than the vent hole or which fails to meet the applicable delay requirement shall be classed defective (see 6.6).

4.3.6 *Impact functioning.* The sample units shall be temperature conditioned at the specified temperature for 6 hours, minimum. To maintain temperature until tested, the units shall each be conditioned in an insulating container with the cover removed from the containers during conditioning. (A low density plastic foam is suggested for container material.) Before removal from the chamber, the cover shall be placed on the container and shall not be removed until the unit is to be assembled into the test equipment. The test shall be performed with the least possible delay. Delay times shall be recorded (see 6.6). Any unit failing to meet the applicable delay requirement shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Packaging for Delay Elements T5E3 or T6E4 shall be in accordance with dwg. 8846862.

5.2 Packing.

5.2.1 *Level A.* Packaging for Delay Elements T5E3 or T6E4 shall be in accordance with dwg. 8846861.

5.3 Marking.

5.3.1 Marking shall be in accordance with dwg. 8846861 or 8846682, as applicable.

5.4 Data card. Data card information shall be prepared for each lot in accordance with Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Type of delay element required.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.

CDCF —Contractor's design responsibility on contractor format.

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OD —Ordnance design.

ODMU —Ordnance design, mandatory for use.

CE —Commercial equipment.

SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation for clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Intended use. Delay elements T6E4 and T5E3 are intended for use with Bomb Fuze, T788E3 (M906).

6.5 Check test for primer deterioration. The contracting officer shall arrange for the contractor to be reimbursed for the expense

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of incurring the performance of the check test. The test shall be conducted at Government expense without cost to the contractor assembling the primer into the delay element, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness on handling, storage, etc., permitted while the primer lot was under jurisdiction of either contractor.

6.6 Test data. Two copies of the test data from each lot are to be forwarded to Commanding Officer, Picatinny Arsenal, Dover, N. J. ATTN: SMUPA–ND4.

6.7 Method of manufacture. Suitability of the igniter and flash and delay composition manufacturing methods shall be determined from the ability of the loaded delay element assembly to meet the requirement specified. After the manufacturing method has been established and placed in production, there shall be no basic change in this method without prior approval of the procuring activity. Records shall be maintained to show the method employed for each lot of loaded and assembled delay elements.

6.8 Process.

6.8.1 *Igniter and flash composition (see 6.7).* The dried mixture of this composition should finally pass through a No. 30 sieve (Specification RR–S–366).

6.8.2 *Granulation of delay composition.* Delay composition should be granulated with a small amount of ethyl alcohol, be forced through a No. 30 sieve and then dried free of alcohol.

6.9 Warning. Both powdered zirconium and the igniter and flash compositions are very sensitive to initiation by the discharge of static electricity, as well as by the application of heat or flame. The igniter and flash composition is also very sensitive to friction and impact forces and has been known to ignite when poured. The principle dangers resulting from the ignition of the dry powders are the intense heat developed and the spattering of the molten slag which is formed upon combustion. However, when either zirconium or igniter and flash composition are wet with quantities of water up to 15 or 20 percent by weight, the reaction becomes explosive in nature due to the reaction between zirconium and water to form hydrogen gas. The hazard is also increased when the igniter and flash composition is wet with an inflammable solvent before initiation. The intense heat of the burning composition vaporizes the solvent and a flash fire results. Operators should be provided with suitable protective covering when handling more than 10 grams of either material. Because of the hazards involved, it is recommended that the container size be limited to approximately one pound.

Custodian:

Army—MU

Preparing activity:

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